CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 16: Loci (Locus and its Constructions)

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Loci in a triangle

1. Statement 1: The point equidistant from three non-collinear points D, E and F is the circumcentre of $\triangle DEF$. Statement 2: The incentre of a triangle is the point where the bisectors of its angles meet. (a) both true (b) both false (c) 1 true, 2 false (d) 1 false, 2 true ICSE 2025, 1 mark

SOLUTION

A point equidistant from the three vertices lies on the perpendicular bisectors of all three sides. That point is the circumcentre, so statement 1 is true.

The angle bisectors meet at the point equidistant from the three sides, which is the incentre, so statement 2 is true.

Answer: (a)

2. The circumcentre of a triangle is the point that is: (a) equidistant from the three sides (b) equidistant from the three vertices (c) the meeting point of the three medians (d) the meeting point of the three altitudes ICSE 2024, 1 mark

SOLUTION

The circumcentre is the centre of the circle through the three vertices, so it is the same distance from each vertex.

Answer: (b)

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2 Construction questions

3. Using ruler and compasses, construct $\angle ABC = 90^\circ$ with $AB = 6$ cm and $BC = 8$ cm. Construct (a) the locus of points equidistant from B and C (b) the locus of points equidistant from A and B (c) the locus of points at the fixed distance OA from O, where O is the point common to (a) and (b) (d) the locus of points equidistant from BA and BC. ICSE 2024, 4 marks

SOLUTION

Draw $BC = 8$ cm, construct a right angle at B and cut off $BA = 6$ cm.

(a) Draw the perpendicular bisector of BC.

(b) Draw the perpendicular bisector of AB.

(c) The two bisectors meet at O. With centre O and radius OA, draw a circle. It passes through A, B and C.

(d) Draw the bisector of $\angle ABC$.

Check: O is the mid-point of the hypotenuse AC, and $AC = \sqrt{6^2 + 8^2} = 10$ cm, so the radius OA should measure 5 cm.

Answer: the loci are two perpendicular bisectors, a circle of radius 5 cm and the bisector of $\angle ABC$

4. Using ruler and compasses, draw a circle of radius 4.5 cm and a chord $AB = 6$ cm. (a) Construct the locus of points equidistant from A and B, and mark the point D where it meets the circle. (b) Join AD, construct the locus of points equidistant from AD and AB, and mark the point C where it meets the circle. (c) Join BC and CD and measure CD. ICSE 2020, 4 marks

SOLUTION

(a) The locus is the perpendicular bisector of AB. It passes through the centre and meets the circle at two points; take D as the point on the far side of the centre from AB.

(b) The locus is the bisector of $\angle DAB$. It meets the circle again at C, which is the mid-point of the arc BD.

(c) Join BC and CD. Because AC bisects $\angle DAB$, the chords BC and CD are equal.

Answer: CD measures about 5.1 cm

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5. Using ruler and compasses, construct a semicircle on the diameter $BC = 7$ cm. Locate the point A on the semicircle that is equidistant from B and C. Complete the cyclic quadrilateral $ABCD$ with D equidistant from AB and BC. Measure $\angle ADC$. ICSE 2019, 4 marks

SOLUTION

Draw $BC = 7$ cm and its perpendicular bisector, meeting BC at O. With centre O and radius 3.5 cm draw the semicircle.

The perpendicular bisector is the locus of points equidistant from B and C. It meets the semicircle at A.

The bisector of $\angle ABC$ is the locus of points equidistant from AB and BC. It meets the semicircle at D. Join AD and DC.

Check: $\angle BAC = 90^\circ$ and $AB = AC$, so $\angle ABC = 45^\circ$. Opposite angles of a cyclic quadrilateral add up to 180°.

Answer: $\angle ADC = 135^\circ$

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